

Development and testing of GaN Power Modules for EV traction inverters.

Dima Novo, Adam Briga, Gregory Bunin, Ilia Bunin, Oren Cohen, Joey Chang, Oleg Dubinsky, Alex Firtel, Yuri Gitelmakher, Nissim Iskhakbayev, Dieter Liesabeths, Mahsa Montazeri, Memo Romero, Evgeny Rozanov, David Shapiro, Lev Stessin, Ray Tsai, Roman Volkov, Valery Veprinsky, Dana Veprinski, Yulia Roiter, Simon Wainwright, VisIC technologies

Dima@visic-tech.com

Abstract

The necessity for affordable long-distance electrical cars is widely recognized. Gallium Nitride (GaN) is an acknowledged semiconductor technology for achieving this target through improved efficiency of the power train inverter. However, to successfully implement GaN in electric vehicle (EV) inverters, the challenges of high-power delivery must be resolved. High power, from a technical perspective, requires the design of high voltage, high current dice, driving in parallel of several dice, and maintaining low thermal resistance between the transistor junction and the coolant liquid of the inverter cold plate. Using power modules instead of discrete devices is an industry mainstream allowing for significantly higher power density. Developing and manufacturing modules with a new semiconductor material, such as GaN, and meeting the aggressive demand of carmakers to reduce cost, weight, and size of power modules while increasing delivered power calls for extraordinary inventiveness of engineers and well-orchestrated communication between packaging, electronics and semiconductor experts. To efficiently integrate power modules in EV inverters, and meet all the requirements stated above, A profound understanding of the trade-offs between different performance parameters and size is essential. This work will demonstrate how these challenges have been addressed in VisIC D3GaN power modules using the methodology of a demonstrator module based on discrete devices and the development of a transfer molded power module, based on bare dice. These modules were developed to showcase the D3GaN technology for EV traction inverters supporting up to 150kW and a battery voltage of 400V.

1 Introduction

In this paper, we will present our D3GaN power module technology demonstrators. The modules are developed to define main tradeoffs, test and showcase the technology for EV traction inverters. D-mode GaN and D3GaN specifically present several benefits that make it highly suitable for applications that require high power, high reliability, and low cost. Fundamentally, fast turn-on and turn-off and very low switching losses of GaN devices improve application efficiency even at relatively low switching frequencies of 20kHz and allow for much higher frequencies when motor technology allows. At the same time, gate reliability is a critical requirement for automotive high-power applications. In the challenge of keeping simultaneously advantages of GaN and reliable operation at high power, D-mode GaN devices show a significant advantage over E-mode GaN devices. We adopted internally developed D3GaN technology, as most suitable for high-power automotive applications, where “D3” stands for Direct Drive D-mode. The gate driver is connected directly to the GaN Gate which allows utilizing the fast turn-on and turn-off times and slew-rate control. D3GaN topology is radically different from cascode topology. First, we present and discuss the technology demonstrator VM022 Half Bridge power module based on D3GaN 8mOhm discrete devices, which was developed to showcase high power operation of 4 parallel devices in a traction inverter application. In this paper, we will present the electrical and mechanical aspects of the module design and the effi-

ciency test results performed on the dyno closed-loop test bench according to the World Harmonized Light Vehicles Test Procedure (WLTP). Second, we will present the production target transfer molded module that is currently in development, showcasing high power density and improved thermal resistance.

2 VM022 D3GaN Power Module design

2.1 Module Overview

Fig. 1 shows the VM022 650V, 2mOhm D3GaN Half Bridge Power module. The key performance parameters are shown in Table I



Figure 1: VM022 Half Bridge side view

TABLE I
VM022 Key Performance Parameters

Parameter	Value	Unit
V_{DS}	650	V
$R_{DS(ON)}$	2	m Ω
Q_G	800	nC
$I_{D,pulse}$	1520	A
I_D	460	A

VM022 is a PCB-based Half Bridge module, with 4 parallel D3GaN 8mOhm discrete devices for high and low sides, a total of 8 GaN dice. The driving board with connectors to a control board is placed on top of the module. Gate drivers are located above the devices to reduce the gate loop length and inductance. The module block diagram is shown in Fig. 2

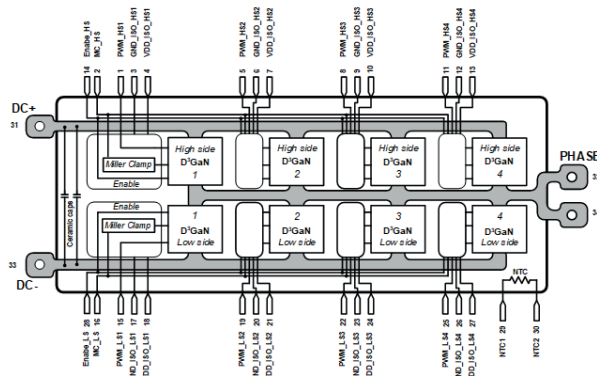


Figure 2: VM022 Block Diagram

To further reduce the power loop inductance decoupling capacitors are integrated on the module PCB.

2.2 Power and Gate loop design considerations

Power and gate loop inductances have a critical impact on system operation. Reducing the parasitic inductances is even more important in applications utilizing GaN devices due to their high switching speeds, leading to high di/dt. Reducing voltage overshoots, minimizing power losses, and EMI. Much attention was put into reducing power and gate loop inductances, both in the power PCB layout, control pin's location, and driver board PCB layout. The measured power loop inductance is approximately 3.9nH. To minimize mismatch in time during turn-on between parallel power devices, special attention was put into the design of the power and driver PCBs, ensuring that the control signals will commute the devices simultaneously. In power loop design we need to make sure that the parasitic inductances per device are equal in value to create even current distribution during switching events. GaN devices have a positive $R_{on}/Temp$ coefficient, i.e. R_{on} increases with an increase in temperature. This effect helps to "naturally" balance current between parallel devices during ON state. Reduction of power loop parasitic inductances is done by 2 methods described in [1],[2]: 1) reducing the area enclosed by the power loop.

2) using field cancellation techniques. In Fig. 3 you can see the power PCB stack-up. In red is the High Frequency current path where (1) & (2) are implemented.

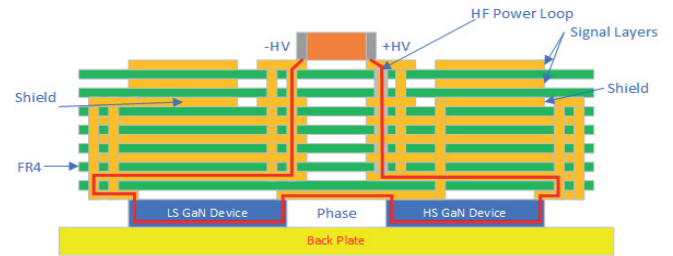


Figure 3: VM022 Power PCB stack-up

2.2.1 Operation of parallel devices

For testing the parallel operation of the GaN devices we have developed a special technique utilizing the LV MOSFET integrated in the D3GaN platform. By measuring the voltage drop on the MOSFET of parallel devices during turn-on, we can verify that the devices turn on simultaneously (current flow begins at the same time). In Fig. 4,5&6 the voltage spike on the parasitic inductance of LV MOSFET during turn-on is shown comparing pairs of the 4 parallel devices. In these waveforms, it is easy to see that current flow starts simultaneously on the devices ($\Delta t=2.807nS$, at its max, on device #1 the farthest from reference device #4) (Blue is the DUT, Gray is the reference device).

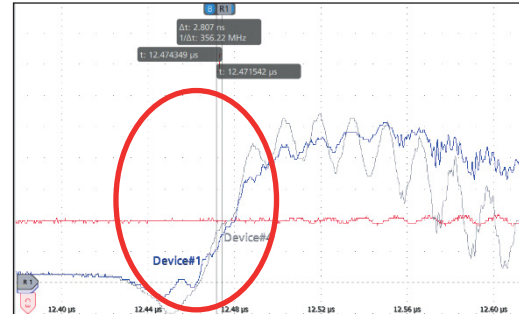


Figure 4: VM022 LV MOSFET or device #1 compared to device #4

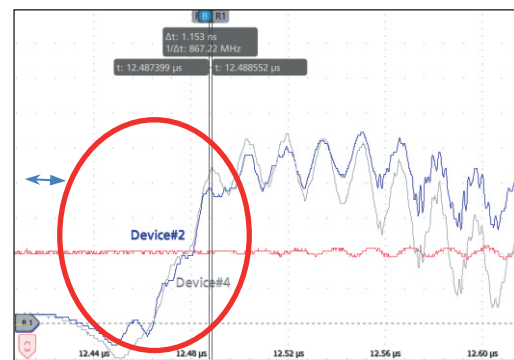
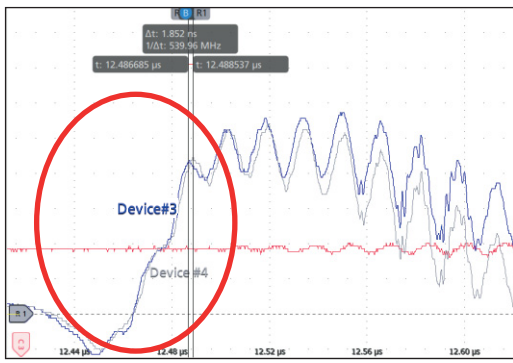


Figure 5: VM022 LV MOSFET or device #2 compared to device #4



Figure

6: VM022 LV MOSFET or device #3 compared to device #4

2.3 Thermal stack

As described in the previous sections and figures, the VM022 power module uses a multilayer power PCB for routing both signal and power traces. Soldered to one side of this PCB are the pre-packaged discrete 8 mOhm D3GaN devices. These packages are electrically isolated via an internal Aluminum Nitride DBC substrate and are then attached to a copper baseplate via a standard soldering process. The assembly, shown in Fig. 7, is considered a typical “baseplate” power module from a thermal standpoint but uses a combination of wirebonding and PCB layers for all interconnections.

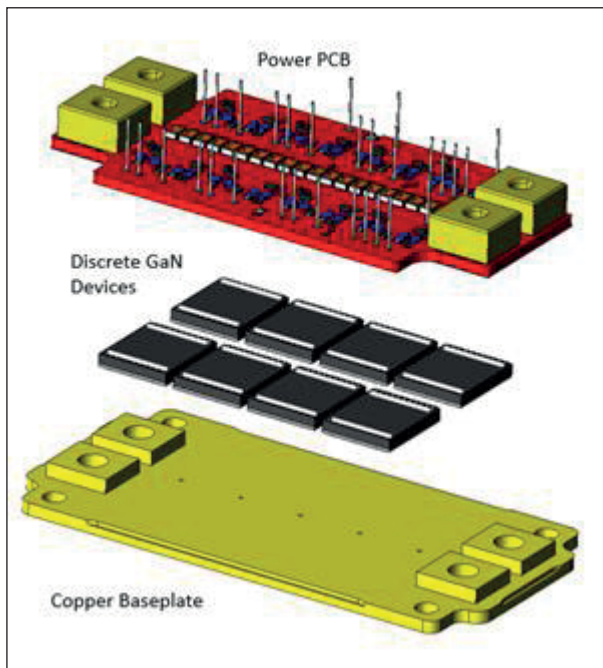


Figure 7: VM022 Power PCB, Discrete devices, and the baseplate

The industry mainstream is to use semiconductor bare die directly attached to the DBC and interconnect through wire/ribbon bonding or clip connection. While this approach is optimal for making mass-production products, it makes the process of tuning, optimization, or debugging of the module design practically impossible. Therefore, we used a different approach which allows tuning of the lay-

out and circuit on a “living” technology demonstrator module. The primary difference is that the pre-packaged discrete GaN devices are first assembled and tested to confirm the operation, then attached both to the power PCB on the terminal side and the baseplate on the DBC side. This provides a conventional thermal path to a cooler (typically liquid cooled), but also a secondary thermal path to the power PCB, where some heat, 10% to 15%, can be dissipated via the thick copper traces, encapsulation materials, and surrounding environment. Fig. 8 depicts the complete vertical structure from die to fluid.

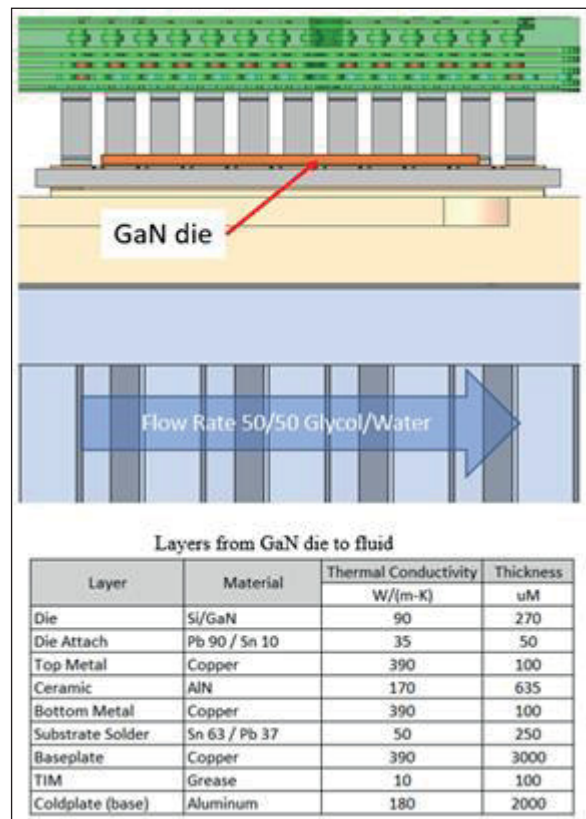


Figure 8 VM022 Thermal stack

With this thermal stack, and with typical fluid flow rates of 8-10 LPM per inverter, we can expect a thermal resistance of device channel to fluid inlet in the range $R_{th, C-FLUID}$ 0.4 to 0.55 °C/W per die, depending on details of the cooler. Some improvements can be made but these must be balanced with minimizing mechanical stresses and other reliability considerations.

3 WLTC test results

WLTC is a cycle test that is part of WLTP [3] and is a globally adopted chassis dynamometer test cycle for the determination of emissions and fuel consumption for ICU or electric efficiency for EV light-duty vehicles. The test was set up in the lab of a major EU car manufacturer. The set-up included a 3-phase inverter power stage (Fig. 9) based on a single VM022 module per phase, a dynamometer system with an EV PMSM machine, a DC power supply, a cooling system, and other auxiliary systems required for the operation and measurement during the experiment.

The whole 3-phase power stage system, including control and measurement equipment, was designed and assembled by the VisIC team.



Figure 9: VM022 3-phase power stage test system

3.1 WLTP tested conditions:

In Fig. 10 we show the full WLTP test conditions (conditions tested are marked in red). In this experiment, we tested up to 350A_{RMS} phase current and 2400rpm (electrical frequency of 160Hz).

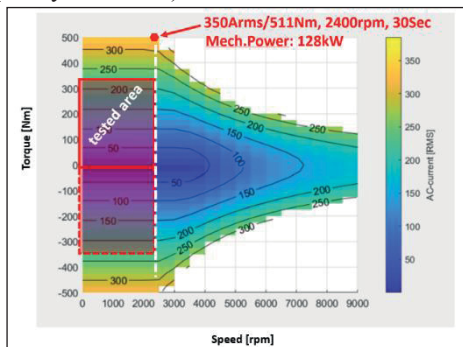


Figure 10: WLTP Current Vs. RPM

Red dot indicates the maximal tested conditions. The red rectangle indicates the area where several data points were collected to build a Torque Vs. Efficiency plot (Fig. 11)

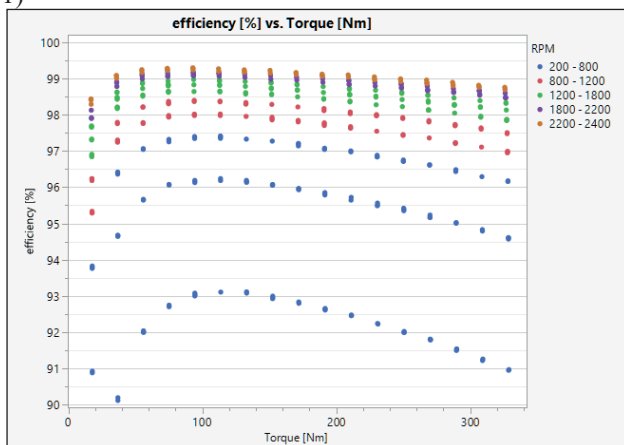


Figure 11 Measured Torque & Efficiency

In this test, we achieved a maximal efficiency of 99.295% at 24.4kW output, 2400RPM, and 92.8Nm.

4 D3GaN Transfer Molded Half-Bridge Module

Following the successful development and testing of the VM022 module VisIC took the next step of developing a transfer-molded Half Bridge module. This module is much more compact: 50x30x4mm [LxWxH]. Power terminals are designed for laser welding to reduce power loop inductance when connected to busbars, and the module is designed for sintering or soldering to a cold plate to reduce thermal resistance. The development is a work in progress and here we're presenting preliminary data.

4.1 Module structure

For optimal thermal efficiency as well as long-term reliability, the VM027 Module structure is designed for a straightforward assembly process and transfer molding. The module is designed for a direct attachment of the Direct Bonded Copper ceramic substrate (DBC) to the cold plate, in this design a baseplate is unnecessary, lowering the cost and improving the thermal performance. Figure 12 below shows an exploded view of this design.

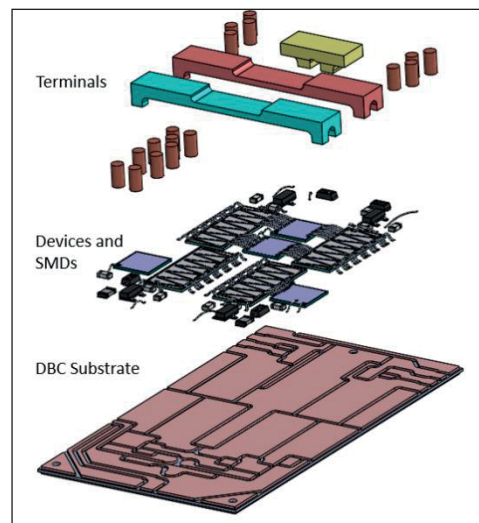


Figure 12: Transfer molded module exploded view

One beneficial feature of this structure is that a typical stamped lead frame is not needed for the power or signal terminals. Instead, low-cost metal injection components are used and placed using standard pick-and-place processes. The novel power terminal geometry provides for low inductance and allows for laser welding to a system-level power connection. The fabricated module after molding is shown in Figure 13.

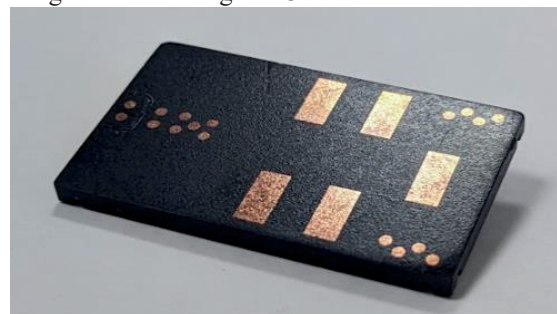


Figure 13: Transfer molded module

4.1.1 Power and Gate loop design considerations

The incentive for minimizing power and gate loop inductance is the same as for the VM022 module. The transfer molded modules present additional constraints: Smaller size and less area for a high cross-section of the power conductors. Cancellation layers are also not practical as readily available DBC has only a single layer of copper. Multi-layer DBCs have been long promised but we were not able to exercise this path. As mentioned above power terminals enable laser welding to busbars, reducing the overall power loop inductance. Control pins are located to ensure a minimal length of control paths. ANSYS Q3D is used for power and gate loop parasitic simulations. Based on the simulations the main power loop inductance is 2.45nH and the gate loop inductance of 6.82nH for the high side gate and 7.53nH for the low side gate.

4.1.2 Paralleling of GaN dice

GaN die statistical V_{th} data as measured on-wafer is shown in Fig. 12. (46,004 dice). Narrow V_{th} spread assists with the paralleling of dice, as the turn-on and turn-off difference at the same gate voltage is low. Real-life operation (as we did with the VM022 module) has not been performed yet.

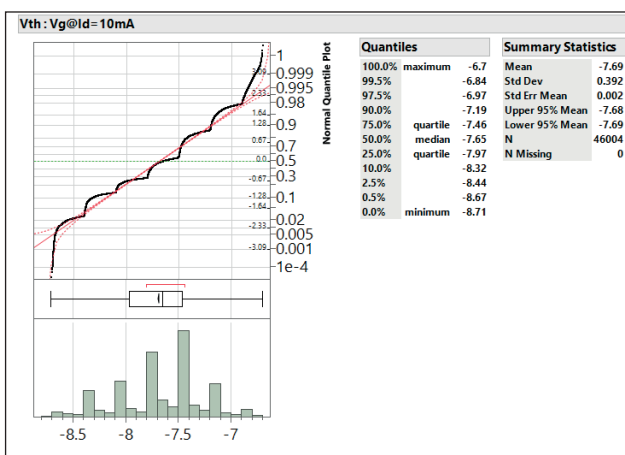


Figure 12: Threshold Voltage Statistics

4.1.3 Thermal stack

This power module is a DBC based transfer molded power module that does not utilize a typical copper baseplate, allowing for direct and thermally efficient attachment to a cooler via solder or large area Ag-sinter bonding. This design has many of the same features as other transfer molded products suitable for automotive applications, including a high-volume production assembly process and cost structure. To maintain an acceptable final flatness of the substrate and to minimize the stresses on the GaN die special attention was given to optimize the mechanical design, as well as the molding process parameters. For the transfer molded module, this has been achieved by careful optimization of DBC layout geometry and metal thicknesses, mechanical design, die attach material, and mold design details. Figure 12 shows this module design. Power terminals are designed for laser welding, which has become the preferred method due to lower costs and the elimination of typical electrical contact resistances.

The substrate is Silicon Nitride (Si_3N_4) due to its superior mechanical strength, compared to AlN, and its increased acceptance in the power electronics industry. Without a baseplate and associated attachment layer, the thermal stack, shown in Figure 13 is further simplified. Consequently, thermal resistance is also lower, with achievable $R_{th\ C-FLUID}$ values between 0.3 and 0.5 $^{\circ}\text{C/W}$ depending on details of the cooler and with flow rates of 8-10 LPM.

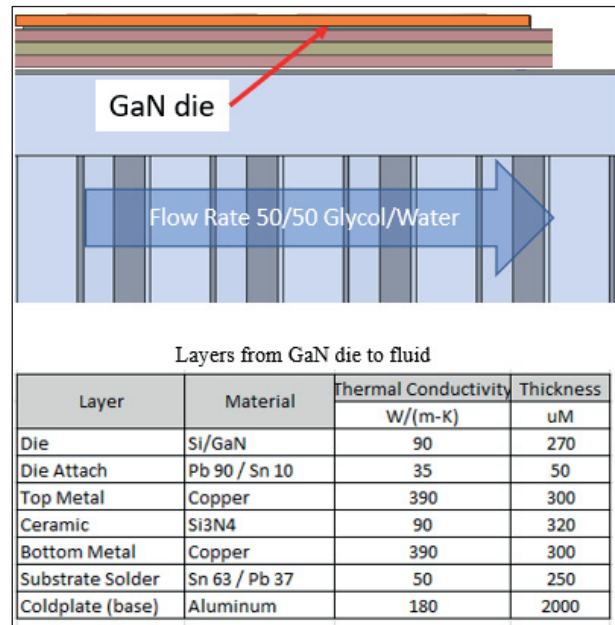


Figure 13: Module thermal stack

The DBC metal thickness in this design is 300uM, much thicker than in the VM022 demonstrator, primarily because this layer must handle the full current load directly on the substrate. A benefit of this approach is more heat spreading, where there is space around the GaN device. The Si_3N_4 ceramic is thinner than AlN to compensate for the lower thermal conductivity value.

4.2 Multi Pulse Test setup & Preliminary Results

For dynamic evaluation of the module, we are using a multi-pulse test. After module fabrication and testing that the module is electrically operational, a multi-pulse test set-up is assembled using a dedicated evaluation board, developed for the module by VisIC team, and providing all the necessary control signals. A 1-GHz high bandwidth oscilloscope and high accuracy optical differential probes are used to measure Drain-Source and Gate signals with minimal distortion. The test set up is shown in Fig. 14. The module is located at the bottom of the EVB.

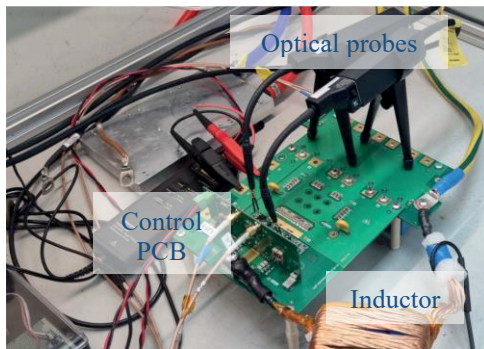


Figure 14: Multi-pulse test setup.

At this early stage of the development, the purpose of the testing is to verify the stability of the switching, evaluate the power loop parasitic inductances, and measure switching energy and additional parameters that are critical for making improvements for the next revision. In Fig. 15 the half bridge mid-point to ground voltage and the coil current were captured. In this test bus voltage is 400V, coil current was up to 200A.

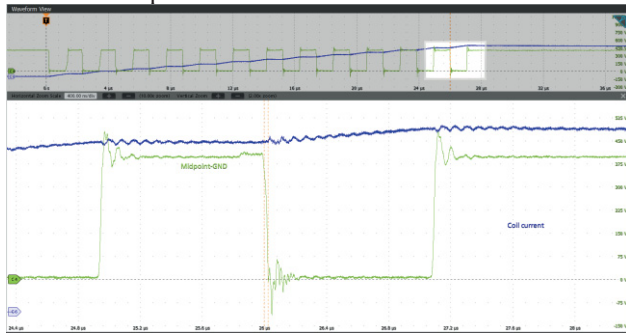


Figure 15: – Transfer molded HB module multi-pulse waveform

The measured power loop inductance is 8nH (including busbars and power capacitor), this measurement indicates a high inductance of the connection between the DC link capacitor and the module.

Switching Energy measurements are next in line, after finishing the initial optimization of the switching behavior.

5 D³GaN Driving Circuit and it's implementation with D³GaN Power Modules.

D³GaN schematic is shown in Fig. 16 and includes: 1) D-mode GaN die, 2) LV MOSFET die, 3) pn Diode. In this example, the GaN and P-MOSFET die are connected in series Source to Source. The pn diode is connected between the GaN Gate and the Drain of the LV MOSFET.

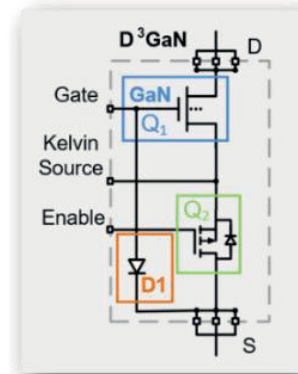


Figure 16:– D³GaN Schematics

The LV MOSFET is in conductive mode all the time driver is present, and not playing a role in the switching of the GaN die. Fig. 17 & 18 show the simplified gate drive and MOSFET enables circuit schematics. The gate of the MOSFET is connected to a dedicated enable circuitry. Enable signal turns the MOSFET to ON state when the auxiliary power supply turns ON and stays ON as long as the auxiliary power is above the Under Voltage Lockout (UVLO) Level. If the auxiliary power supply voltage level drops below the UVLO, the MOSFET turns OFF and allows for a safe turn-off of the GaN die.

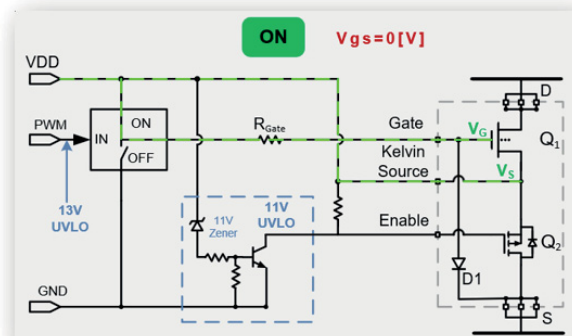


Figure 17: – D³GaN Driving Circuit Schematics – ON state

Driving of the D³GaN

Standard off-the-shelf gate drivers are used for driving a D³GaN and are connected to the GaN Gate (sizing of the gate driver is done in the same way as for SiC or IGBT power devices). There is no need in inverting or changing the driver input PWM signal. The difference in the D³GaN driving circuit is that the GaN die Kelvin Source is connected to the auxiliary power supply (VDD) instead of GND. This allows for switching the GaN Gate between VDD and GND potentials, while GaN Source is hardwired to VDD.

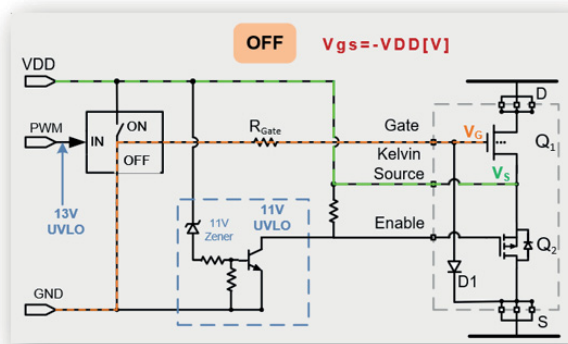


Figure 18: – D3GaN Driving Circuit Schematics – OFF stat.

For Turn OFF the GaN $V_{GS} = GND - VDD = -VDD$ (Fig. 18), and ON when $V_{GS} = VDD - VDD = 0$.

For implementing the D³GaN inside a module, it is required to have the GaN and Si MOSFET on the substrate for cooling, while all the other components can be implemented on the Gate drive PCB.

D³GaN utilizes the benefits of a more robust D-mode GaN device with highly reliable gate structure without the added switching losses of the Si MOSFET.

6 Literature

- [1] Eric Persson: Optimizing PCB Layout for HV GaN Power Transistors. IEEE Power Electronics Magazine, Vol. 10, No.2 June 2023.
- [2] Juncheng(Lucas) Lu: Paralleling GaN E-HEMTs in 10kW-100kW Systems, 2017 IEEE Applied Power Electronics Conference and Exposition (APEC).
- [3] <https://dieselnet.com/standards/cycles/wltp.php>